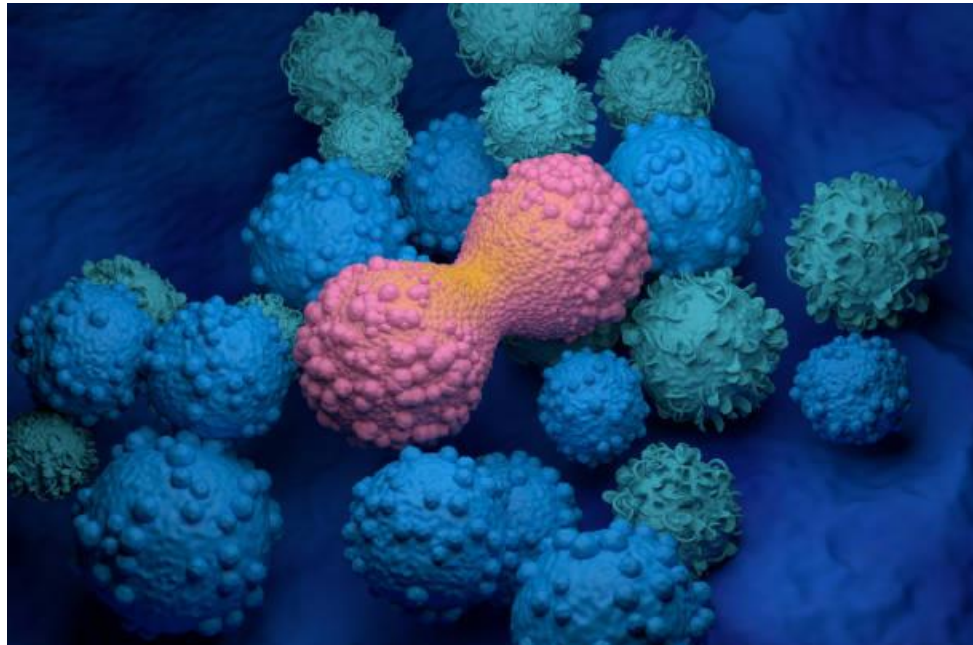


# **What is cancer? How do we name them?**

**CCC419**

**Spring 2026**

**Dr. Anindita Chakrabarty**

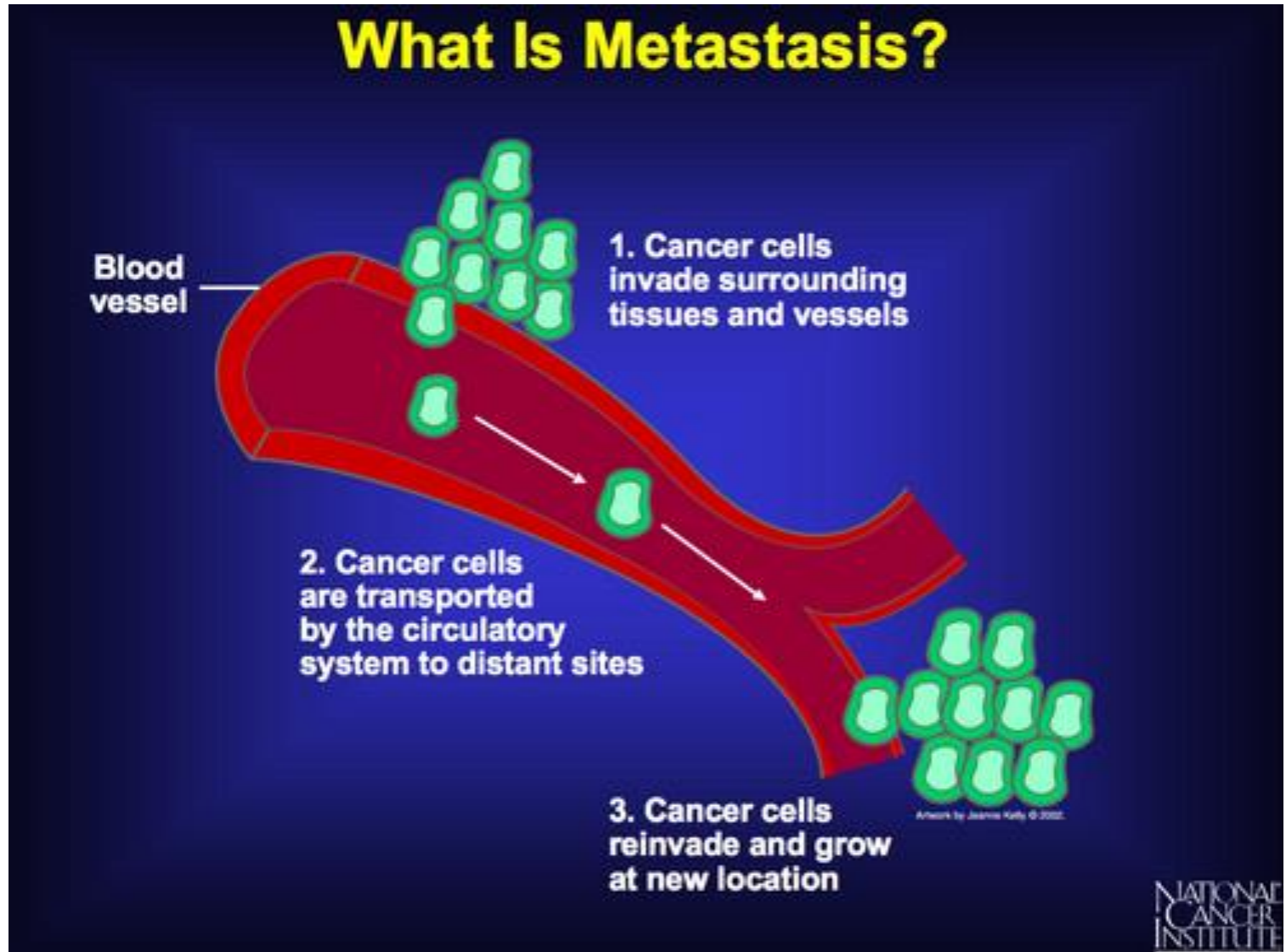


# What is cancer? What kind of disease is it?

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- ✓ Group of related diseases in which cells grow without restriction, do not die even when damaged, and spread into nearby sites or distant tissues and organs of the body through circulation.
  - ✓ More than 100 different diseases\* are called cancer
1. *Tissue/organ where the disease starts\**
  2. *The causative agent (cigarette smoke, sun UV exposure)\**
  3. *Molecular mechanisms (genetic mutation caused by 2)\**
  4. *Treatment (how lung versus skin cancer is treated)\**

# Cancer spread=metastasis

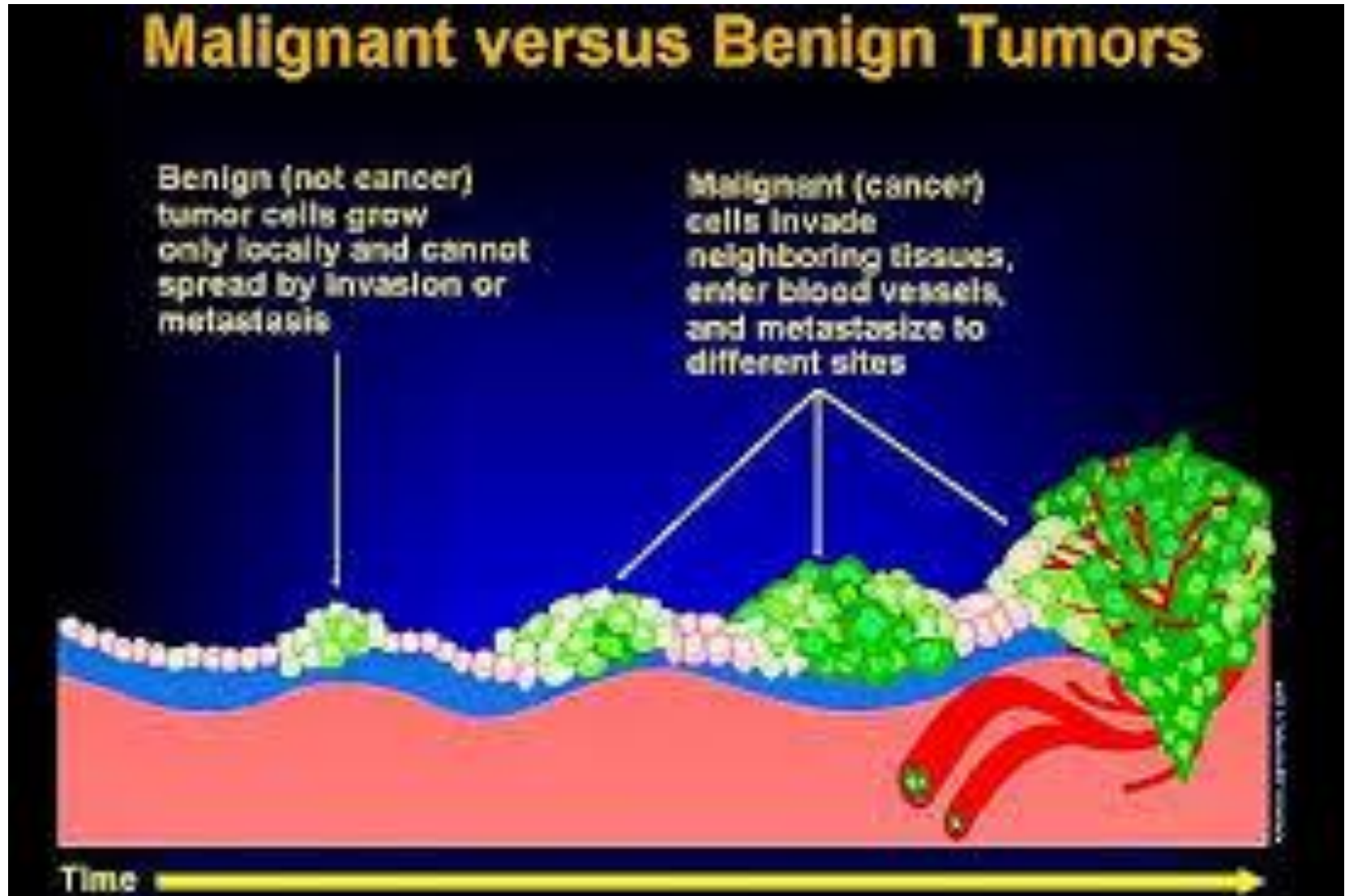


# Benign Vs. Malignant Tumor- Which one is cancerous and life-threatening?

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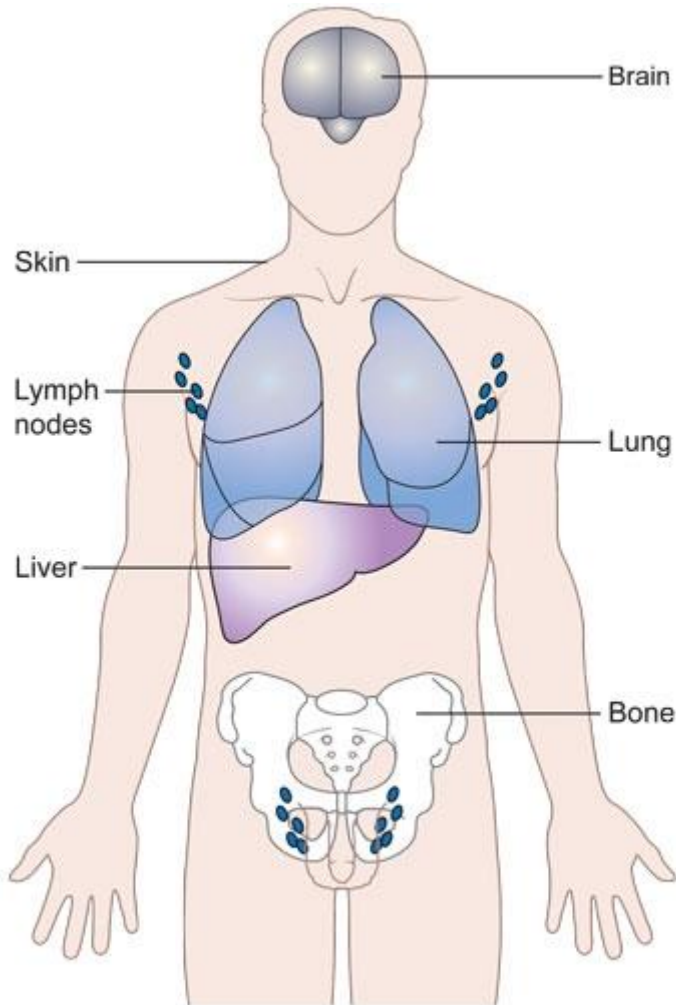
- ✓ Benign tumors do not spread throughout the body
- ✓ Malignant tumors break through the tissue boundaries, invade and metastasize (??)
- ✓ *Malignant tumors are life-threatening because they are physical obstructions and can invade other organs, impairing their function*
- ✓ *Malignant tumors also compete with healthy tissues for nutrients and oxygen*
- ✓ *Benign brain tumors can be life-threatening*

# Benign tumor versus malignant tumor (cancer)



# Where do cancers spread to? Any organ or specific organ?

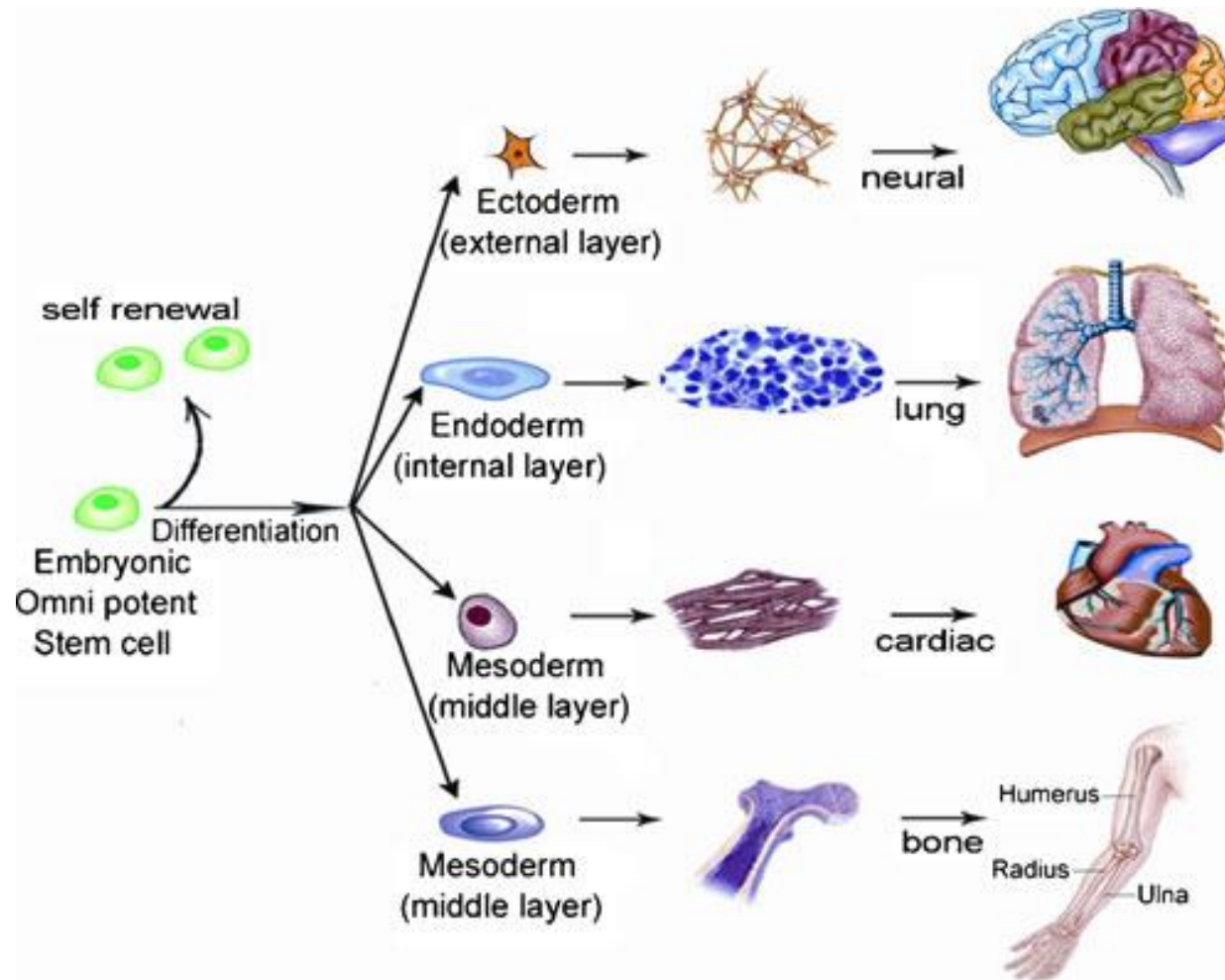
✓ Lung, Liver, Lymph nodes, Bones, Brain, Skin



- Not all cancers spread to all organs
- There is selectivity
- Seed and soil hypothesis
- Why is this information important?



# Different organs come from different cell types



# Cancers can be categorized into different groups; cell type is important

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## 1. **Carcinomas:** Cancers of epithelial cells

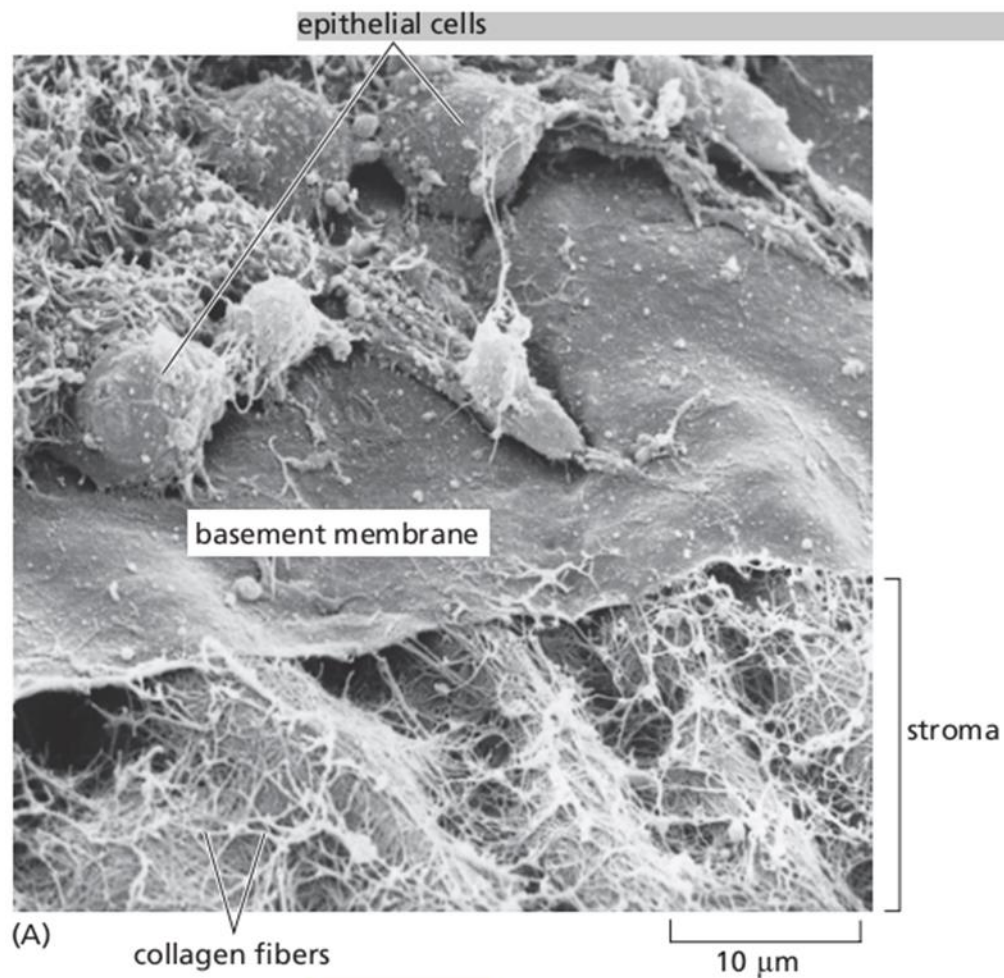
- **Adenocarcinomas:** Cancers of tissues forming glands (lung, stomach, uterus, breast)
- **Squamous cell carcinomas:** Cancers of protective layers (skin, oral cavity)
- 90% of human cancers are **carcinomas**, responsible for more than 80% of cancer deaths

## 2. **Sarcomas:** Cancers of mesodermal cells (bone, muscle, fat, cartilage)

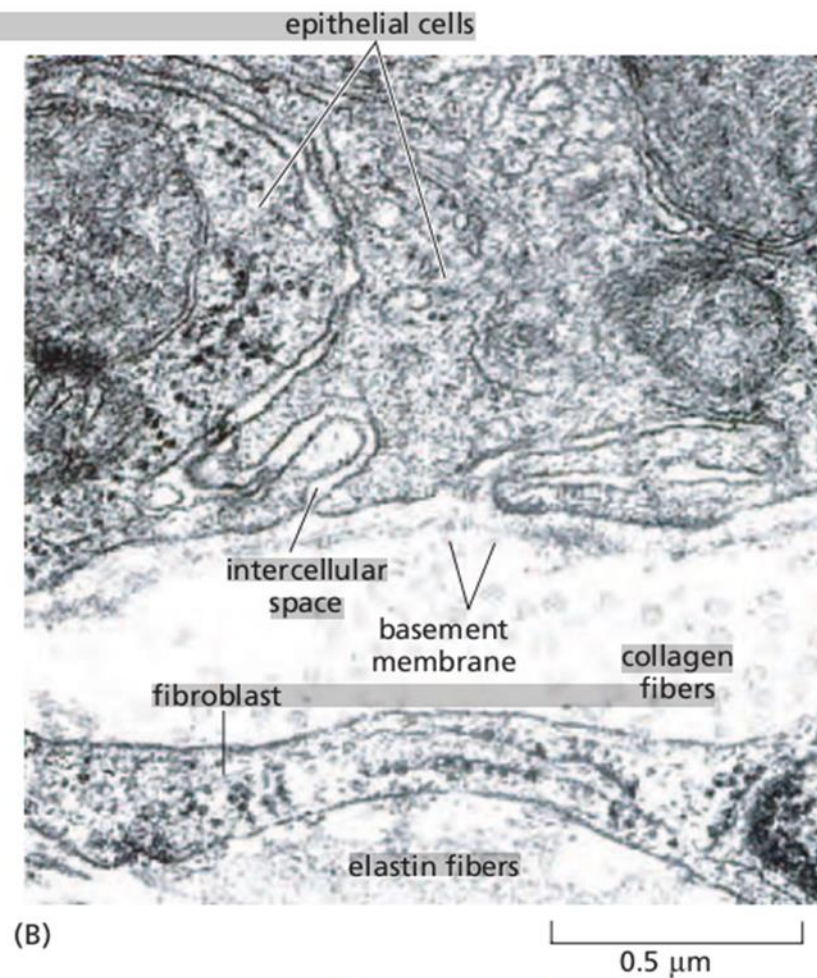
## 3. **Blood** cancers

## 4. **Brain** cancers





**Cornea**



**Trachea**

**Common tumors from epithelial cells are those of the GI tract-mouth, esophagus, stomach, small and large intestines, as well as of mammary gland, skin, pancreas, lung, liver, ovary, uterus, prostate, gallbladder and urinary bladder**

**Table 2.1** Carcinomas

| Tissue sites of more common types of adenocarcinoma                                             | Tissue sites of more common types of squamous cell carcinoma                | Other types of carcinoma                                                                                                                                          |
|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| lung<br>colon<br>breast<br>pancreas<br>stomach<br>esophagus<br>prostate<br>endometrium<br>ovary | skin<br>nasal cavity<br>oropharynx<br>larynx<br>lung<br>esophagus<br>cervix | small-cell lung carcinoma<br>large-cell lung carcinoma<br>hepatocellular carcinoma<br>renal cell carcinoma<br>transitional-cell carcinoma<br>(of urinary bladder) |

# Tumors that are not carcinomas

- ✓ Derived from middle layer/connective tissues, known as **sarcomas** (1% of all tumors)
- ✓ **Sarcomas can be different types**

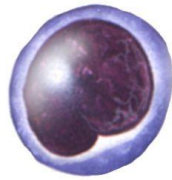
Table 2.2 Various types of more common sarcomas

| Type of tumor                 | Presumed cell lineage of founding cell      |
|-------------------------------|---------------------------------------------|
| Osteosarcoma                  | osteoblast (bone-forming cell)              |
| Liposarcoma                   | adipocyte (fat cell)                        |
| Leiomyosarcoma                | smooth muscle cell (e.g., in gut)           |
| Rhabdomyosarcoma              | striated/skeletal muscle cell               |
| Malignant fibrous histiocyoma | adipocyte/muscle cell                       |
| Fibrosarcoma                  | fibroblast (connective tissue cell)         |
| Angiosarcoma                  | endothelial cells (lining of blood vessels) |
| Chondrosarcoma                | chondrocyte (cartilage-forming cell)        |

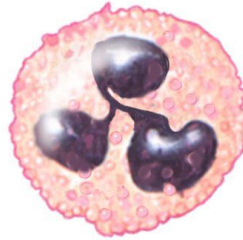
## Blood Cells



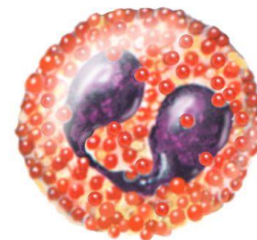
**Monocyte**



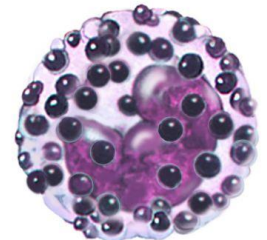
**Lymphocyte**



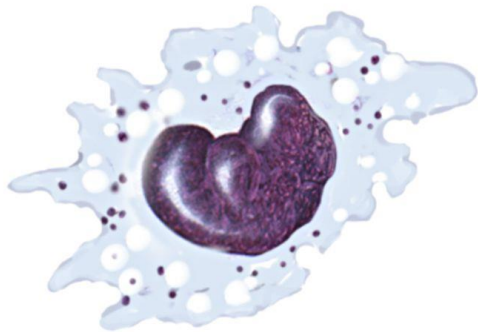
**Neutrophil**



**Eosinophil**



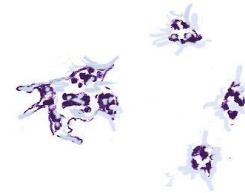
**Basophil**



**Macrophage**



**Erythrocyte**



**Platelets**



# Blood cancers: leukemia, lymphoma and multiple myeloma

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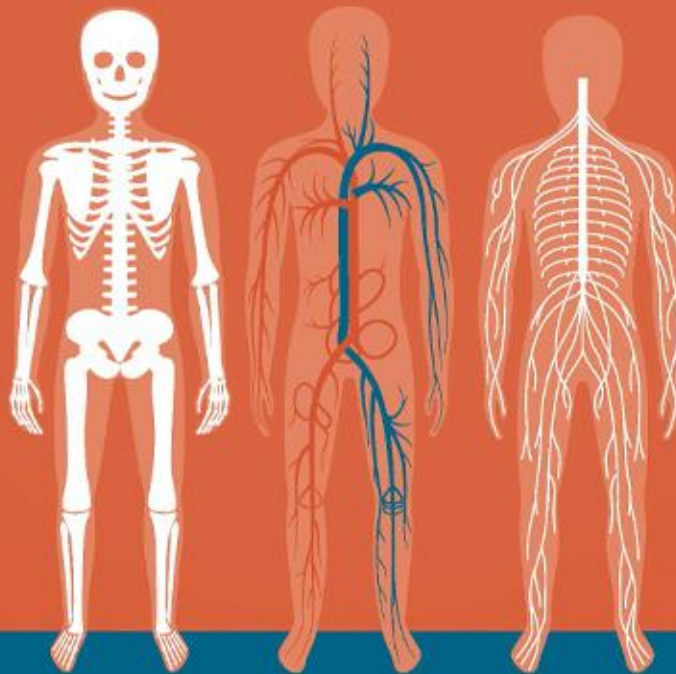
- ✓ Another type is formed from *blood-forming tissues including immune cells*
- ✓ What are blood forming cells? RBC, WBC, T and B cells, platelets
- ✓ *Leukemia: malignant derivatives of blood cells that move freely in circulation and are non-pigmented*
- ✓ *Lymphoma: tumors of lymphoid origin (T and B cells) that aggregate to form solid tumor masses frequently found in lymph nodes-accounts for about 7% of cancer related mortality*
- ✓ *T=from thymus*
- ✓ *B-from bone marrow*

# LEUKEMIA VS. LYMPHOMA

Lymphoma and leukemia, often called liquid cancers or blood cancers, share a common origin—lymphocytes, or white blood cells of the immune system.

**Leukemia** is cancer of the blood cells that usually starts in bone marrow and often travels through the bloodstream.

BONE MARROW



**Lymphoma** is cancer of the lymphatic system that usually originates in lymph nodes or, sometimes, the spleen.

LYMPH NODES





**Table 2.3** Various types of more common hematopoietic malignancies

Acute lymphocytic leukemia (ALL)  
Acute myelogenous leukemia (AML)  
Chronic myelogenous leukemia (CML)  
Chronic lymphocytic leukemia (CLL)  
Multiple myeloma (MM)  
Non-Hodgkin's lymphoma<sup>a</sup> (NHL)  
Hodgkin's lymphoma (HL)

<sup>a</sup>The non-Hodgkin's lymphoma types, also known as lymphocytic lymphomas, can be placed in as many as 15–20 distinct subcategories, depending upon classification system.

# Brain tumors

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- ✓ Third major group of non-epithelial tumors arise from *cells of central and peripheral nervous system-neuroectodermal tumors*
- ✓ Included are gliomas, glioblastomas, neuroblastomas, schwannomas, medulloblastomas
- ✓ *Comprises of 1.3% of diagnosed cancers, but responsible for 2.5% of cancer deaths*

**Table 2.4** Various types of neuroectodermal malignancies

| Name of tumor              | Lineage of founding cell                      |
|----------------------------|-----------------------------------------------|
| Glioblastoma multiforme    | highly progressed astrocytoma                 |
| Astrocytoma                | astrocyte (type of glial cell) <sup>a</sup>   |
| Meningioma                 | arachnoidal cells of meninges <sup>b</sup>    |
| Schwannoma                 | Schwann cell around axons <sup>c</sup>        |
| Retinoblastoma             | cone cell in retina <sup>d</sup>              |
| Neuroblastoma <sup>e</sup> | cells of peripheral nervous system            |
| Ependymoma                 | cells lining ventricles of brain <sup>f</sup> |
| Oligodendroglioma          | oligodendrocyte covering axons <sup>g</sup>   |
| Medulloblastoma            | granular cells of cerebellum <sup>h</sup>     |

<sup>a</sup>Nonneuronal cell of central nervous system that supports neurons.

<sup>b</sup>Membranous covering of brain.

<sup>c</sup>Constructs insulating myelin sheath around axons in peripheral nervous system.

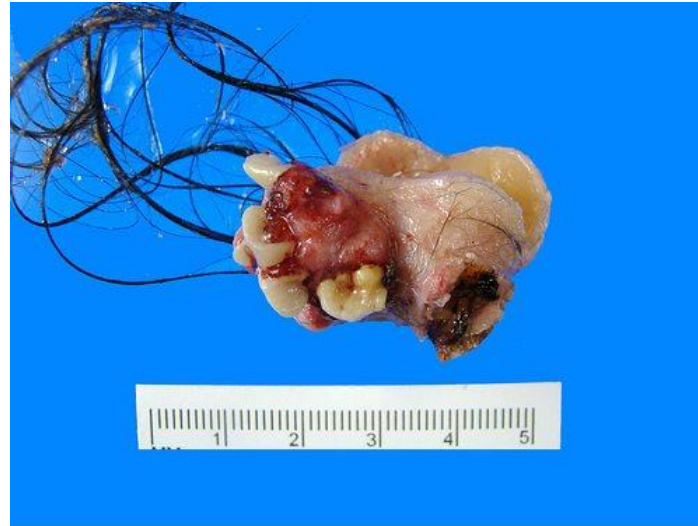
<sup>d</sup>Photosensor for color vision during daylight.

<sup>e</sup>These tumors arise from cells of the sympathetic nervous system.

<sup>f</sup>Fluid-filled cavities in brain.

<sup>g</sup>Similar to Schwann cells but in brain.

<sup>h</sup>Cells of the lower level of cerebellar cortex (for example, see Figure 2.9B).



**A Teratoma, meaning ‘monster tumor’, is a rare growth inside the human body that can have hair and teeth.**

# Tumors of Undefined Origin

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- ✓ Two to four percent of all tumors lose all the status of the tissue-of-origin
- ✓ Tumors are called ***anaplastic***
- ✓ *Anaplastic tumors are often categorized as the cancer of unknown primary*

